

# EDGE CASE TESTING & VALIDATION

## AI-Powered Career Opportunity Assistant

### 1. Introduction

This document presents the edge case testing performed on the AI-powered workflow system. The objective of this testing is to validate the system’s ability to handle real-world variations in email inputs, including incomplete data, irrelevant content, and multiple opportunities.

Each test case includes:

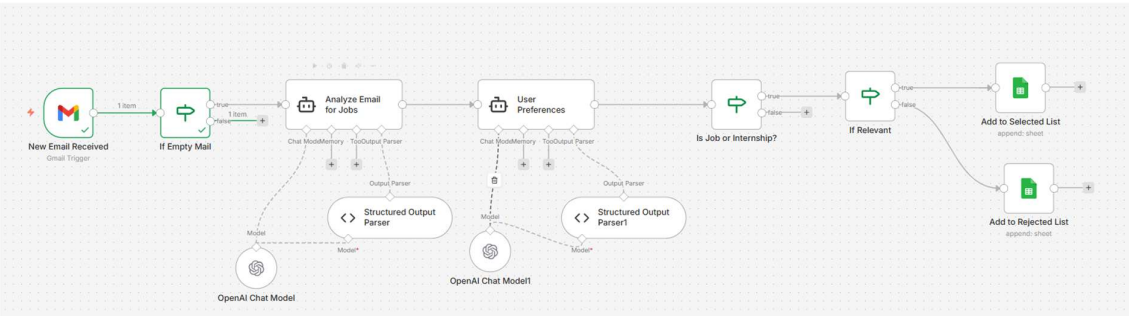
- Input Email
- Workflow Execution
- System Output
- Observation

### 2. Edge Case 1: Empty / No Content Email

Input:



Workflow:



Output:

- No data processed
- Workflow stops at initial validation step

✅ **Observation:**

The system correctly identifies empty or invalid email content and prevents further processing, ensuring no unnecessary AI calls or incorrect data entries.

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### 3. Edge Case 2: Non-Job Email

Input:

Invitation to a party - reg

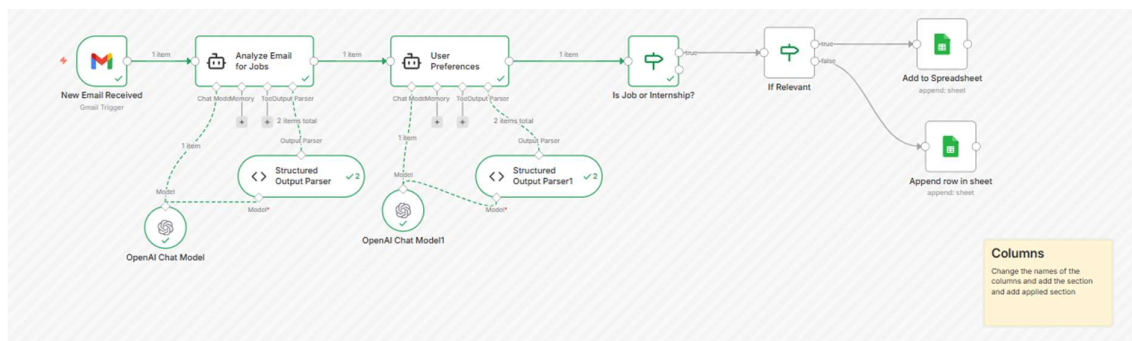


**TRIPTI KUMARI** <tripti27kumari@gmail.com>  
to newtripti925@gmail.com

11:22 AM (5 hours ago)

Hi, Tripti this is to inform you that our company is hosting an event related to new graduates and learn from their journey and how did they crack companies. We would like to invite you to our esteemed party as a guest speaker. looking forward to learning from you.  
Event name : Our seekers  
Date: 12 april 2026  
venue : college campus  
time: 12 pm - 5pm

Workflow:



Output:

Is Job or Internship?

INPUT Schema Table JSON Parameters Settings

Conditions

fx: `{{ $json.output.isJobOrInternship }}` is true

Options

Convert types where required

Options

OUTPUT Schema Table JSON

True Branch False Branch (1 item)

output

isJobOrInternship: false

company: unknown

role: Guest Speaker

salary: unknown

link: unknown

platformOrUser: unknown

deadline: 12 April 2026

offerType: Unknown

isRelevant: false

reason: This is not a job or internship offer but an invitation for a Guest Speaker role. It lacks employer, application link, salary, and is outside the user's preferred roles (Software Developer, Data Analyst) and minimum salary requirement. It appears to be a generic/event invitation rather than a real employment opportunity, so it is not relevant.

- isJobOrInternship = false
- Email ignored

#### ✓ Observation:

The system successfully filters out **non-job-related emails**, preventing irrelevant data from entering the pipeline.

## 4. Edge Case 3: Job Email (Not Relevant to User)

Input:

(no subject)

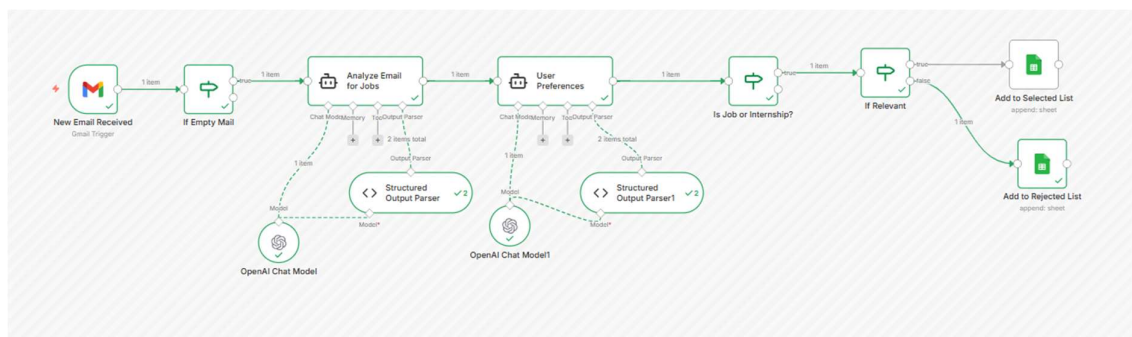
TRIPTI KUMARI <tripti27kumari@gmail.com>  
to me

4:22 PM (41 minutes ago) ☆ 🗨 ⌂

Hello Tripti this is to inform you that , we require you to fill the form for our upcoming opportunity for hiring campus.  
Hiring event : [Narula Hiring Campus](#)  
Link : <https://www.jobcliff.com/job-detail/510>  
salary : 8LPA  
Skills: [sales](#), marketing, consulting  
Requirements: CSE graduate , 2026 batch  
Role: [sales](#) role, team leader  
deadline: 15th april 2026

Hi, I am interested. I have registered. Ok, I will do it.

Workflow:



Output:

The screenshot shows a data processing tool interface with three main sections: INPUT, PARAMETERS, and OUTPUT.

**INPUT:** A table with one item. The 'output' column contains a JSON object with the following fields: `isJobOrInternship`: true, `company`: Narula, `role`: sales role, team leader, `salary`: 8LPA, `link`: https://www.jobcliff.com/job-detail/510, `platformOrUser`: Narula Hiring Campus, `deadline`: 15th april 2026, `offerType`: Job, `isRelevant`: false, and `reason`: Role is a sales/team-leader position which does not match the user's stated preferred roles (Software Developer, Data Analyst) or technical skillset. Although salary (8 LPA) exceeds the user's minimum and graduation year (2026) makes them eligible, the domain mismatch makes this opportunity not relevant. The listing is specific and includes an application link (not a generic email).

**PARAMETERS:** A section titled 'Values to Send' with three rows: `Date` (fx: {{ new Date().toLocaleDateString() }}), `isRelevant` (fx: {{ \$json.output.isRelevant }}), and `platformOrUser` (fx: {{ \$json.output.platformOrUser }}).

**OUTPUT:** A table with one item. The 'output' column contains a JSON object with the following fields: `isJobOrInternship`: true, `company`: Narula, `role`: sales role, team leader, `salary`: 8LPA, `link`: https://www.jobcliff.com/job-detail/510, `platformOrUser`: Narula Hiring Campus, `offerType`: Job, `isRelevant`: false, `reason`: Role is a sales/team-leader position which does not match the user's stated preferred roles (Software Developer, Data Analyst) or technical skillset. Although salary (8 LPA) exceeds the user's minimum and graduation year (2026) makes them eligible, the domain mismatch makes this opportunity not relevant. The listing is specific and includes an application link (not a generic email), `match_score`: 30, and `Date`: 4/8/2026.

- isRelevant = false
- Match score: Low

✔ Observation:

The system evaluates the opportunity against the user profile and correctly rejects roles that **do not align with user preferences**, ensuring personalized filtering.

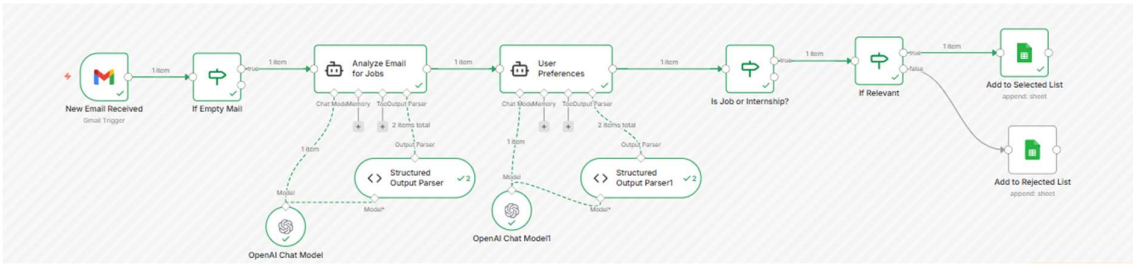
5. Edge Case 4: Job Email (Relevant to User)

Input:

The screenshot shows an email invitation from Genpact to Tripti Kumari. The email is titled "Tripti Kumari, you're invited to walk-in" and includes the following details:

- Genpact** (3.5 (41798 Reviews))
- Kolkata Walkin\_Fresher For Transaction risk investigator role 8 April**
- Durgapur, Kolkata, Howrah**
- 0 - 0 years**
- Not Disclosed**
- In office**
- Email support, Email Process, Domestic Process, Chat, Web Chat, Semi Voice, Customer Service, Customer**

Workflow:



Output:

The screenshot shows the 'Add to Selected List' output window. The 'OUTPUT' tab is active, displaying a table with one item. The table has columns for 'platform', 'deadline', 'is\_job\_or\_internship', 'salary', 'offerType', 'link', 'Company', and 'Summary'. The data for the single item is as follows:

| platform   | deadline    | is_job_or_internship | salary        | offerType | link                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Company | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|-------------|----------------------|---------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Naukri.com | 08-Apr-2026 | true                 | Not Disclosed | Job       | <a href="http://my.naukri.com/AL/ResdexRMJMail/aliid/%7B%7D/redirectParam/applyBroadcastMail?xz=1.0.66&amp;sajeid=1&amp;showLB=1&amp;f=060426015686&amp;mid=060426015686&amp;rdxMsgId=060426015686&amp;ssageId=060426015686&amp;src=saje&amp;applySrc=M/R&amp;uld=a4f3a88ead372f9d2f47008dbe0f31f&amp;vCd=5132608&amp;utm_campaign=broadcast_rmj&amp;utm_medium=email&amp;utm_source=imlnterested">http://my.naukri.com/AL/ResdexRMJMail/aliid/%7B%7D/redirectParam/applyBroadcastMail?xz=1.0.66&amp;sajeid=1&amp;showLB=1&amp;f=060426015686&amp;mid=060426015686&amp;rdxMsgId=060426015686&amp;ssageId=060426015686&amp;src=saje&amp;applySrc=M/R&amp;uld=a4f3a88ead372f9d2f47008dbe0f31f&amp;vCd=5132608&amp;utm_campaign=broadcast_rmj&amp;utm_medium=email&amp;utm_source=imlnterested</a> | Genpact | Relevant: The role involves transaction risk/fraud investigation which aligns well with the candidate's hands-on fraud-detection and data-analytics experience (built fraud detection on 6.3M transactions, ML, SQL, Power BI). Graduation year (2026) fits the job timeline and the candidate prefers Data Analyst roles, which overlap with this position. Caveat: salary is not disclosed (candidate's minimum is 4 LPA) — confirm compensation before applying. |

- isRelevant = true
- Match score: High

✓ Observation:

The system successfully identifies and prioritizes opportunities that match the user's:

- Preferred roles
- Skillset
- Career stage

6. Output Storage (Google Sheets)

Sheet 1 (for selected Job/Internship)

|   | A        | B           | C             | D          | E                             | F             | G          | H           | I                                                                         | J                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | K | L |
|---|----------|-------------|---------------|------------|-------------------------------|---------------|------------|-------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| 1 | DATE     | Platform    | Company       | OfferType  | Role                          | Salary        | Deadline   | Match_score | Link                                                                      | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |
| 2 | 4/8/2026 | Other       | XYZ Analytics | Internship | Data Analyst Intern           | ₹15000/month  | 2026-05-10 | 95          | <a href="https://xyz.com/apply">https://xyz.com/apply</a>                 | Good fit: Data Analyst role aligns with user's preferences and strong skills (Python, SQL, Power BI, ML), suitable for a 2026 graduate and internships are accepted regardless of stipend.                                                                                                                                                                                                                                                                           |   |   |
| 3 | 4/8/2026 | Other       | XYZ Analytics | Internship | Data Analyst Intern           | ₹15000/month  | 2026-05-10 | 95          | <a href="https://xyz.com/apply">https://xyz.com/apply</a>                 | Good fit: Data Analyst role aligns with user's preferences and strong skills (Python, SQL, Power BI, ML), suitable for a 2026 graduate and internships are accepted regardless of stipend.                                                                                                                                                                                                                                                                           |   |   |
| 4 | 4/8/2026 | Naukari.com | Genpact       | Job        | Transaction Risk Investigator | Not Disclosed | 2026-04-08 | 80          | <a href="60428015686&amp;messageId=604">60428015686&amp;messageId=604</a> | Relevant. The role involves transaction risk/fraud investigation which aligns well with the candidate's hands-on fraud-detection and data-analytics experience (built fraud detection on 6.3M transactions, ML, SQL, Power BI). Graduation year (2026) fits the job timeline, and the candidate prefers Data Analyst roles, which overlap with this position. Caveat: salary is not disclosed (candidate's minimum is 4 LPA) — confirm compensation before applying. |   |   |

Sheet 2 (for rejected Job/internship)

|   | A        | B                    | C          | D             | E                       | F             | G           | H                                                           | I                                                                                                                                                                                                                                                                                                                                                                                                 | J | K |
|---|----------|----------------------|------------|---------------|-------------------------|---------------|-------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| 1 | Date     | Platform             | OfferType  | Company       | Role                    | Salary        | Match_score | link                                                        | Summary                                                                                                                                                                                                                                                                                                                                                                                           |   |   |
| 2 |          |                      |            |               |                         |               |             |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |
| 3 | 4/8/2026 | Nanula Hiring Campus | Job        | Nanula        | sales role, team leader | 8LPA          | 30          | <a href="www.jobcliff.com/job-d">www.jobcliff.com/job-d</a> | Role is a sales/team-leader position which does not match the user's stated preferred roles (Software Developer, Data Analyst) or technical skillset. Although salary (8 LPA) exceeds the user's minimum and graduation year (2026) makes them eligible, the domain mismatch makes this opportunity not relevant. The listing is specific and includes an application link (not a generic email). |   |   |
| 4 | 4/8/2026 | Other                | Internship | XYZ Analytics |                         | ₹15,000/month | 25          | <a href="https://xyz.com/apply">https://xyz.com/apply</a>   | Role not specified/unclear (reject). Internship is suitable for a 2026 grad but salary (~₹1.8 LPA) is below the user's 4 LPA preference.                                                                                                                                                                                                                                                          |   |   |
| 5 | 4/8/2026 | Other                | Job        | Nanula        |                         | 8 LPA         | 30          | <a href="www.jobcliff.com/job-d">www.jobcliff.com/job-d</a> | Role is unspecified/unclear in the job data, so the listing is rejected despite salary and skill alignment.                                                                                                                                                                                                                                                                                       |   |   |

 **Stored Fields:**

- Company
- Role
- Salary
- Link
- Platform
- Deadline
- Offer Type
- Match Score
- Relevance

 **Observation:**

All processed data is stored in a **structured format**, enabling tracking, filtering, and future analysis.

## 7. Testing Limitation

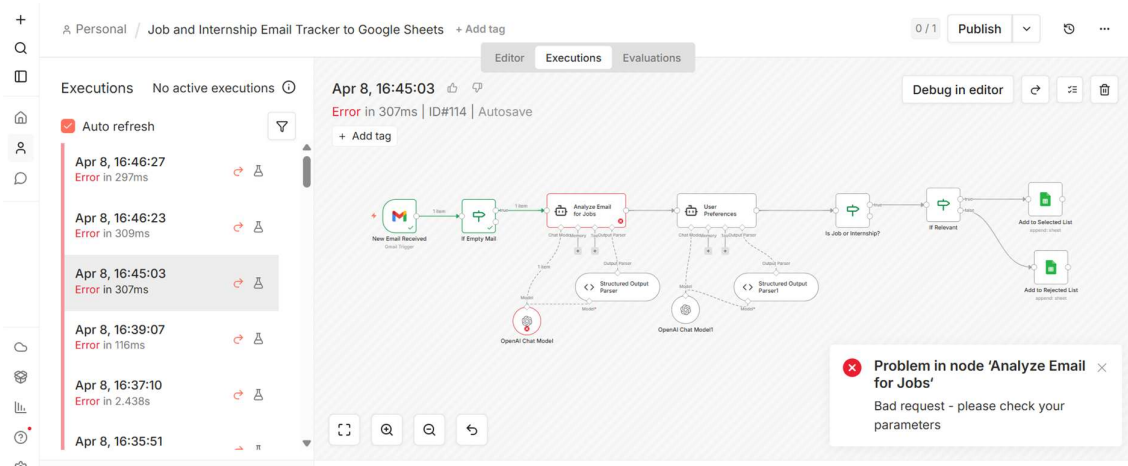
During testing, the workflow was executed **multiple times (100+ executions)** to validate different scenarios and edge cases.

Due to this extensive testing:

- The allocated **API usage credits were exhausted**
- Further live executions could not be performed

### Note:

The examples provided above represent key validated scenarios. Additional cases are logically supported by the implemented system rules but could not be executed due to API limitations.



Personal / Job and Internship Email Tracker to Google Sheets + Add tag

0 / 1 Publish

Editor Executions Evaluations

Apr 8, 16:45:03 Error in 307ms | ID#114 | Autosave + Add tag

Debug in editor

Auto refresh

Apr 8, 16:46:27 Error in 297ms

Apr 8, 16:46:23 Error in 309ms

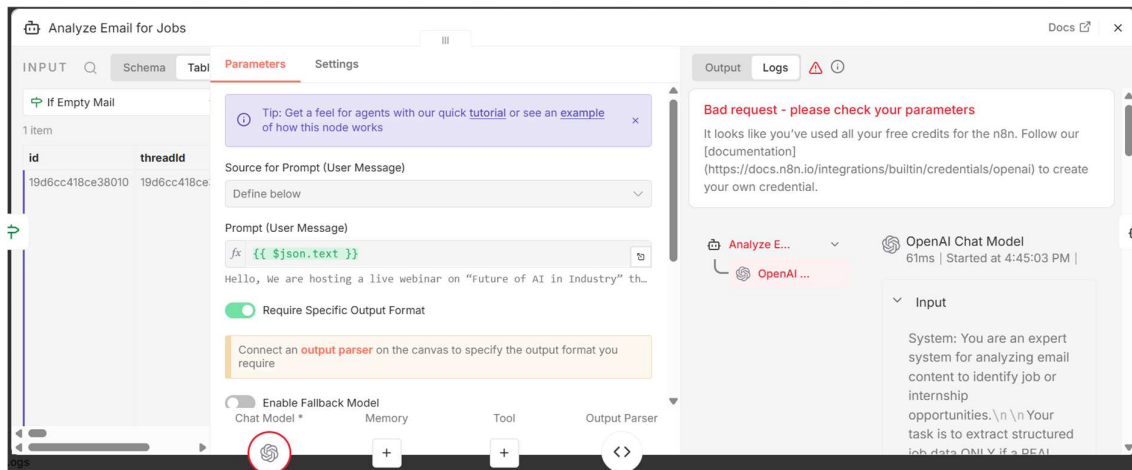
Apr 8, 16:45:03 Error in 307ms

Apr 8, 16:39:07 Error in 116ms

Apr 8, 16:37:10 Error in 2.438s

Apr 8, 16:35:51

Problem in node 'Analyze Email for Jobs' Bad request - please check your parameters



Analyze Email for Jobs

INPUT Schema Tab Parameters Settings

Tip: Get a feel for agents with our quick tutorial or see an example of how this node works

Source for Prompt (User Message) Define below

Prompt (User Message) f: {{ \$json.text }} Hello, We are hosting a live webinar on "Future of AI in Industry" th...

Require Specific Output Format

Connect an output parser on the canvas to specify the output format you require

Enable Fallback Model Chat Model \* Memory Tool Output Parser

Output Logs

Bad request - please check your parameters

It looks like you've used all your free credits for the n8n. Follow our [documentation] (https://docs.n8n.io/integrations/builtin/credentials/openai) to create your own credential.

Analyze E... OpenAI Chat Model 61ms | Started at 4:45:03 PM |

Input

System: You are an expert system for analyzing email content to identify job or internship opportunities.\n\nYour task is to extract structured job data ONLY if a REAL